Supplementary Material for "Physics-Disentangled Adaptive Surrogate-Assisted Optimization for Lifecycle Carbon-Integrated Microgrid Planning"

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S.1. Overview

This supplementary document provides extended mathematical formulations, comprehensive parameters, and in-depth analytical results referenced in the main manuscript. Specifically, it includes the complete nomenclature (Section S.2), the detailed mathematical formulations of the lower-level dispatch model constraints (Section S.3), the derivations for lifecycle financial conversions (Section S.4), the detailed techno-economic parameters and algorithmic hyperparameters (Section S.5), and the SHapley Additive exPlanations (SHAP) physical interpretability analysis of the proposed surrogate model (Section S.6).

S.2. Nomenclature

The complete list of sets, indices, parameters, and decision variables utilized in both the main manuscript and this supplementary document is detailed in Table S1.

S.3. Detailed Mathematical Formulation of the Lower-Level Dispatch Model

For any arbitrary scenario $s \in \mathcal{S}$, the daily comprehensive operational cost $J_{op,s}$ is minimized. While the penalty costs ($C_{penalty}$) and power balance equations are detailed in the main manuscript, the remaining constituent components are rigorously formulated as follows.

A. Baseline Operational Cost (C_{base})

The baseline operational cost aggregates fuel consumption, start-up wear of diesel generators (DGs), net grid interaction costs, and variable operation and maintenance (O&M) costs associated with energy storage systems (ESS).

$$C_{base} = C_{fuel} + C_{start} + C_{grid} + C_{om,bat} \tag{S1}$$

The fuel and start-up costs for diesel generators are calculated as:

$$C_{fuel} = \sum_{t\in\mathcal{T}} \sum_{m\in\mathcal{M}_{dg}} \eta_{fuel,m} P_{dg,m}(t) \pi_{diesel} \Delta t \tag{S2}$$

$$C_{start} = \sum_{t\in\mathcal{T}} \sum_{m\in\mathcal{M}_{dg}} N_{start,m}(t) \pi_{start,m} \tag{S3}$$

The net grid interaction cost reflects energy trading with the main utility grid, incorporating time-of-use (TOU) electricity prices and a sales discount factor γ_{sell}:

$$C_{grid} = \sum_{t\in\mathcal{T}} \left[P_{grid}^{buy}(t)\pi_{elec}(t) - P_{grid}^{sell}(t)\pi_{elec}(t)\gamma_{sell} \right] \Delta t \tag{S4}$$

The variable O&M cost for energy storage evaluates the dynamic usage throughput:

$$C_{om,bat} = \sum_{t\in\mathcal{T}} \sum_{m\in\mathcal{M}_{ess}} P_{dis,m}(t)\pi_{om,m}\Delta t \tag{S5}$$

B. Carbon Tax Cost (C_{tax})

The model imposes a carbon tax on direct operational carbon emissions generated by the system (excluding embodied carbon, which is processed in the upper-level lifecycle model):

$$C_{tax} = \lambda_{tax} \left(E_{dg} + E_{grid} \right) \tag{S6}$$

$$E_{dg} = \sum_{t\in\mathcal{T}} \sum_{m\in\mathcal{M}_{dg}} P_{dg,m}(t)\eta_{fuel,m}\mu_{diesel}\Delta t \tag{S7}$$

$$E_{grid} = \sum_{t\in\mathcal{T}} P_{grid}^{buy}(t)\mu_{grid}\Delta t \tag{S8}$$

C. Operational Smoothing Penalty (C_{smooth})

To mitigate the degradation of energy storage devices, a power smoothing penalty term suppresses high-frequency charging and discharging fluctuations:

$$C_{smooth} = \lambda_{smooth} \sum_{t\in\mathcal{T}} |P_{net}^{ess}(t) - P_{net}^{ess}(t-1)| \tag{S9}$$

$$P_{net}^{ess}(t) = P_{dis}(t) - P_{ch}(t) \tag{S10}$$

D. Physical Constraints

In addition to the fundamental power balance constraint detailed in the main manuscript, the operational variables satisfy the following physical constraints.

1. Operational Logic of Diesel Generators: The number of operating units $N_{on,m}(t)$ at time t is strictly bounded by the total installed capacity N_m determined by the upper-level configuration $\mathbf{x}$:

$$0 \le N_{on,m}(t) \le N_m, \quad N_{on,m}(t) \in \{0, 1, \ldots, N_m\} \tag{S11}$$

To prevent operation in low-efficiency regions, a minimum load ratio constraint α_{dg} is imposed. The total output power $P_{dg,m}(t)$ is bounded by:

$$\alpha_{dg} Cap_m N_{on,m}(t) \le P_{dg,m}(t) \le Cap_m N_{on,m}(t) \tag{S12}$$

The DG start-up logic, utilized to calculate the start-up cost, is mathematically tracked by an auxiliary integer variable $N_{start,m}(t)$:

$$N_{start,m}(t) \ge N_{on,m}(t) - N_{on,m}(t-1) \tag{S13}$$

$$N_{start,m}(t) \ge 0, \quad N_{start,m}(t) \in \mathbb{Z} \tag{S14}$$

TABLE S1
NOMENCLATURE

Sets and Indices			
$\mathcal{S}$	Set of representative scenarios (s)	$\mathcal{T}$	Set of time steps in daily dispatch (t)
$\mathcal{M}$	Set of candidate device types (m)		
Parameters and Constants			
r	Discount rate for lifecycle cost (%)	T_{proj}	Project lifetime (Year)
$Price_m^{inv}$	Unit investment cost (CNY/unit)	$Price_m^{rep}$	Unit replacement cost (CNY/unit)
$Price_m^{fix}$	Annual fixed O&M cost (CNY/unit/yr)	π_{diesel}	Fuel price for diesel generators (CNY/L)
π_{elec}	Grid electricity price (CNY/kWh)	$\pi_{start,m}$	Start-up cost for diesel Gen. (CNY/start)
$\pi_{om,m}$	Variable O&M cost for ESS (CNY/kWh)	λ_{shed}	Load shedding penalty (CNY/kWh)
λ_{curt}	Renewable curtailment penalty (CNY/kWh)	λ_{tax}	Carbon tax rate (CNY/kg)
λ_{smooth}	Battery health depreciation cost (CNY/kW)	λ_{AP}	PVIFA factor (-)
ω_s	Weight of scenario s (-)	Cap_m	Rated capacity of device m (kW)
L_m	Design lifespan of device m (Year)	$\eta_{fuel,m}$	Fuel consumption rate (L/kWh)
η	Charging/Discharging efficiency (%)	R_{max}	Max ramp rate for ESS (kW/h)
P_{grid}^{max}	Grid interaction power limit (kW)	α_{dg}	Min load ratio for diesel Gen. (%)
Δt	Time interval of dispatch (h)	ξ_m^{LCA}	LCA carbon factor (kg/kW)
μ_{diesel}	Diesel combustion emission factor (kg/L)	μ_{grid}	Grid emission factor (kg/kWh)
Decision and Operation Variables			
$\mathbf{x}$	Device configuration vector (-)	N_m	Number of installed units (-)
LCC	Total lifecycle cost (F_1, CNY)	LCE	Total lifecycle carbon emissions (F_2, kg)
Y_{cost}	Annualized operational cost (CNY/yr)	Y_{carb}	Annualized carbon emissions (kg/yr)
E_{shed}^{yr}	Annualized load shedding energy (kWh)	E_{curt}^{yr}	Annualized curtailment energy (kWh)
$J_{op,s}$	Daily operational cost for scenario s (CNY)	P_{dg}	Output power of diesel generators (kW)
P_{grid}^{buy}	Power purchased from grid (kW)	P_{grid}^{sell}	Power sold to grid (kW)
$P_{ch/dis}$	ESS charging/discharging power (kW)	P_{shed}	Load shedding power (kW)
P_{curt}	Renewable power curtailment (kW)	$E(t)$	Energy stored in battery (kWh)
E_{dg}	Diesel carbon emissions (kg)	E_{grid}	Grid purchase carbon emissions (kg)
$N_{on,m}$	Number of running diesel units (-)	$N_{start,m}$	Number of started diesel units (-)
u_{grid}	Grid interaction binary status (0/1)		

2. Energy Storage Dynamics and Ramping: The inter-temporal evolution of the energy state (State of Charge) considers the charging and discharging efficiency η:

$$E(t+1) = E(t) + \left(P_{ch}(t)\eta - \frac{P_{dis}(t)}{\eta} \right) \Delta t \tag{S15}$$

The continuous charging/discharging power transition is subjected to a maximum ramp rate limit R_{max}:

$$|P_{net}^{ess}(t) - P_{net}^{ess}(t-1)| \leq R_{max}\Delta t \tag{S16}$$

3. Grid Interaction Limits: Constrained by the physical capacity of the transmission line at the Point of Common Coupling, net power flows cannot exceed P_{max}^{grid}. Furthermore, purchasing and selling power are mutually exclusive actions, enforced by the binary variable $u_{grid}(t)$:

$$0 \leq P_{grid}^{buy}(t) \leq P_{max}^{grid} u_{grid}(t) \tag{S17}$$

$$0 \leq P_{grid}^{sell}(t) \leq P_{max}^{grid}(1 - u_{grid}(t)) \tag{S18}$$

$$u_{grid}(t) \in \{0, 1\}, \quad \forall t \in \mathcal{T} \tag{S19}$$

E. Annualized Metric Extrapolation

To feed the continuous base cost and base emission data back to the upper-level lifecycle model and the active learning surrogate, the daily operational simulation results are extrapolated into annualized metrics using scenario probabilities ω_s:

$$Y_{cost}(\mathbf{y}^*) = \sum_{s \in \mathcal{S}} \omega_s J_{op,s}(\mathbf{y}_s^*) \tag{S20}$$

$$Y_{carb}(\mathbf{y}^*) = \sum_{s \in \mathcal{S}} \omega_s \sum_{t \in \mathcal{T}} (E_{dg}(t) + E_{grid}(t)) \tag{S21}$$

S.4. DERIVATIONS OF LIFECYCLE FINANCIAL CONVERSIONS

The total lifecycle cost objective (F_1) in the upper-level planning model aggregates multiple capital flows over the project horizon. The rigorous mathematical formulations for discounting these future cash flows to their present values are detailed below.

1. Initial Investment Cost (C_{inv}):
The initial investment represents the Capital Expenditure (CAPEX) incurred during the initial stage of project construction, calculated as the sum of unit costs multiplied by installed quantities:

$$C_{inv} = \sum_{m \in \mathcal{M}} N_m Price_m^{inv} \tag{S22}$$

2. Equipment Replacement Cost (C_{rep}):
Since the operational lifespan L_m of critical devices (e.g.,

batteries) is typically shorter than the total project planning horizon T_{proj}, multiple replacements occur throughout the lifecycle. These future replacement expenditures are discounted to their present value:

$$C_{rep} = \sum_{m \in \mathcal{M}} \sum_{y \in Y_{rep,m}} N_m Price_m^{rep} (1+r)^{-y} \tag{S23}$$

where $Y_{rep,m} = \{kL_m \mid k \in \mathbb{Z}^+, 1 \leq kL_m < T_{proj}\}$ mathematically defines the set of years in which equipment replacements occur.

3. Salvage Value Recovery ($C_{salvage}$):
At the end of the project lifetime T_{proj}, the residual value of the remaining equipment is calculated based on the linear depreciation method and discounted back to the base year:

$$C_{salvage} = \sum_{m \in \mathcal{M}} N_m Price_m^{rep} \frac{L_m - T_{rem,m}}{L_m (1+r)^{T_{proj}}} \tag{S24}$$

where $T_{rem,m} = T_{proj} \pmod{L_m}$ denotes the used service life of the recently replaced device at the end of the project.

4. Annualized Cost Present Value Conversions (C_{fix} and C_{op}^{total}):
The present value interest factor of annuity (PVIFA), denoted as λ_{AP}, converts an annualized recurring cost into its equivalent present value over the project lifetime T_{proj} given a discount rate r:

$$\lambda_{AP} = \frac{1-(1+r)^{-T_{proj}}}{r} \tag{S25}$$

This factor is rigorously applied to calculate the total present value of the fixed operation and maintenance costs (C_{fix}) across all installed capacities, as well as the comprehensive present value of the operational expenditures (C_{op}^{total}) returned from the lower-level dispatch simulation:

$$C_{fix} = \sum_{m \in \mathcal{M}} \left(N_m Price_m^{fix} \right) \lambda_{AP} \tag{S26}$$

$$C_{op}^{total} = Y_{cost}(\mathbf{y}^*) \lambda_{AP} \tag{S27}$$

S.5. Data Processing and System Parameters

A. Scenario Reduction Verification

To control computational dimensionality while preserving the physical fidelity of meteorological and load fluctuations, a K-means clustering algorithm combined with an extreme condition screening mechanism is applied to the 8760-hour historical dataset. Five operating days representing extreme physical conditions are forcibly extracted to ensure boundary safety, and 15 typical seasonal days are generated through clustering. As illustrated in Fig. S1, this reduction strictly retains the statistical characteristics and physical fluctuation boundaries of the original annual time series.

B. Detailed Experimental Parameters

The microgrid planning case study relies on rigorous techno-economic parameters for candidate devices, including their capital expenditures, replacement costs, operation and maintenance costs, and lifecycle assessment carbon factors.

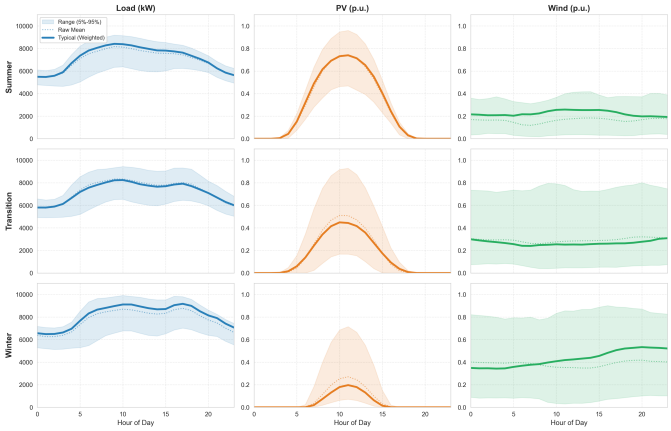

Fig. S1. Visual verification of the scenario reduction method. The shaded areas represent the statistical boundaries of the original annual data, while the solid lines depict the extracted typical and extreme scenarios.

The detailed specifications are provided in Table S2. Furthermore, the global simulation constraints and the algorithmic hyperparameters for the active learning and NSGA-II optimization are summarized in Table S3 and Table S4, respectively.

S.6. Detailed Hyperparameter Configurations for Benchmark Surrogate Models

To ensure the rigorousness and reproducibility of the surrogate model performance verification, the detailed hyperparameter configurations for both the proposed physics-disentangled surrogate and the benchmark models are meticulously documented in Table S5. All models are implemented using Python 3.9.13 and standardized machine learning libraries (*scikit-learn*, *LightGBM*, and *XGBoost*).

Furthermore, to ensure fair comparison and stable convergence, specific data preprocessing mechanisms are applied according to the mathematical characteristics of each algorithm:

- **Data Standardization**: For continuous distance-based and gradient-based models, including Deep Neural Network (DNN), Support Vector Regression (SVR), and Kriging (Gaussian Process), both the input features ($\mathbf{X}$) and the regression targets ($\mathbf{Y}$) are standardized to zero mean and unit variance using the *StandardScaler*. Tree-based models (LightGBM, XGBoost, Random Forest) bypass this step due to their inherent scale invariance.
- **Computational Intractability Mitigation**: SVR and Kriging models exhibit $\mathcal{O}(N^3)$ time complexity during training, making them computationally intractable on large-scale datasets. Therefore, a maximum training sample limit ($N_{limit} = 1000$) is strictly imposed for these two models via a random downsampling mechanism to prevent memory overflow and algorithm freezing, simulating their practical deployment limits in large-scale iterative optimization.

TABLE S2
DETAILED TECHNICAL AND ECONOMIC PARAMETERS OF CANDIDATE DEVICES

ID	Desc.	Size (kW/kWh)	Life (Yr)	Inv. (kCNY)	Rep. (kCNY)	Fix. O&M (CNY)	Var. (C/kWh)	LCA (kg/kW)	Tech. Specs & Constraints	Note
1. Photovoltaic (PV)										
PV-A	Poly-Si	100 / –	25	250	0	1,200	0	1,150	$\eta = 16.5\%$	Low cost, large area
PV-B	Mono-Si	100 / –	25	280	0	1,500	0	1,350	$\eta = 20.5\%$	High cost-performance
PV-C	N-Bifacial	100 / –	25	310	0	1,800	0	1,450	$\eta = 22.5\%$	High albedo suitability
2. Wind Turbine (WT)										
WT-A	Low-Spd	500 / –	20	2,500	2,000	40k	0	850	$v_{in/out} = 3/20$ m/s	Cut-in/out speed
WT-B	Medium	1,500 / –	20	6,000	4,800	80k	0	800	$v_{in/out} = 3.5/25$ m/s	Microgrid mainstay
WT-C	Large	2,000 / –	20	7,800	6,200	100k	0	780	$v_{in/out} = 4/25$ m/s	Low embodied carbon
3. Energy Storage System (ESS)										
BAT-A	Power-2C	250 / 500	12	650	520	6,500	0.05	150	$\eta = 95\%$	Freq. regulation
BAT-B	Energy-4C	250 / 1k	12	1,100	880	11k	0.04	130	$\eta = 94\%$	Peak shaving std.
BAT-C	Long-6C	500 / 3k	12	3,000	2,400	30k	0.03	120	$\eta = 92\%$	Energy shifting
4. Diesel Generator (DG)										
DG-A	Backup	500 / –	15	600	480	15k	0.45	120	Fuel: 0.24 L/kWh; Start: 50¥ Ramp: 10%/min; T: 1h/1h	Min Load 50%
DG-B	Prime	1,000 / –	15	1,100	880	25k	0.40	110	Fuel: 0.23 L/kWh; Start: 80¥ Ramp: 10%/min; T: 2h/2h	Min Load 100%
DG-C	Main	2,000 / –	15	2,000	1,600	45k	0.35	100	Fuel: 0.22 L/kWh; Start: 150¥ Ramp: 7.5%/min; T: 3h/2h	Min Load 150%
5. Inverter (INV)										
INV-G	General	100 / –	10	20	18	200	0	60	$\eta = 98\%$	1:1 config w/ PV

Note: 1. **Inv.** = Investment Cost (Unit: 1,000 CNY); **Rep.** = Replacement Cost (Unit: 1,000 CNY). 2. **Fix. O&M** values marked with "k" denote thousands. **Var.** = Variable O&M Cost (Unit: CNY/kWh). 3. **Tech. Specs**: For DG, parameters format is "Fuel Rate; Start Cost; Ramp Rate; Min Up/Down Time".

TABLE S3
GLOBAL SIMULATION PARAMETERS

Parameter	Symbol	Value (Unit)
Economic Parameters		
Project Lifetime	T_{proj}	25 (Year)
Discount Rate	r	6% (-)
Diesel Price	π_{diesel}	7.5 (CNY/L)
Load Shedding Penalty	λ_{shed}	20.0 (CNY/kWh)
Curtailment Penalty	λ_{curt}	0.15 (CNY/kWh)
Grid Price Ratio	γ_{sell}	0.3 (-)
Carbon Tax Rate	λ_{tax}	0.1 (CNY/kg)
Environmental Parameters		
Grid Emission Factor	μ_{grid}	0.58 (kg/kWh)
Diesel Emission Factor	μ_{diesel}	2.63 (kg/L)
Physical Constraints		
Max Grid Interaction	P_{grid}^{max}	5000 (kW)
DG Min Load Ratio	α_{dg}	20% (-)
Smoothing Penalty	λ_{smooth}	0.001 (-)
Decision Space		
Max PV Units	N_{PV}^{max}	60 (-)
Max Wind Units	N_{WT}^{max}	6 (-)
Max Battery Units	N_{BAT}^{max}	40 (-)
Max Diesel Units	N_{DG}^{max}	5 (-)

TABLE S4
SUMMARY OF KEY ALGORITHM HYPERPARAMETERS

Parameter	Symbol	Value
Data Processing		
Initial LHS Sample Size	N_{LHS}	5000
Number of Typical Days	K	15
Number of Extreme Days	E	5
Active Learning		
Initial Training Samples	N_{init}	200
Max Active Iterations	I_{max}	15
Query Batch Size	N_{batch}	100
Committee Size	k	5
Exploitation Ratio	ρ	0.3
Decay Rate Coefficient	γ	0.1
Base Weight (Shedding)	α	1.5
Dynamic Gain (Shedding)	β	1.0
Bi-level Optimization		
Population Size	N_{pop}	100
Max Generations	N_{gen}	150
Crossover Probability	p_c	0.9
Mutation Probability	p_m	0.1
Oracle Correction Interval	T_{correct}	10
Independent Runs	N_{runs}	31

S.7. PHYSICAL INTERPRETABILITY OF THE SURROGATE MODEL (SHAP ANALYSIS)

To rigorously verify that the feature extraction mechanism of the proposed surrogate architecture aligns with the underlying energy flow logic—rather than merely capturing statistical coincidences—a SHapley Additive exPlanations (SHAP) attribution analysis is conducted on the operational cost sub-model.

TABLE S5
HYPERPARAMETER SETTINGS FOR THE PROPOSED AND BENCHMARK SURROGATE MODELS

Model	Algorithm Core	Hyperparameter Configurations
Proposed	LightGBM (Sparse Targets)	`objective` = *tweedie*, `tweedie_variance_power` = 1.5, `n_estimators` = 500
	LightGBM (Smooth Targets)	`objective` = *regression* (MAE loss), `n_estimators` = 500
XGBoost	eXtreme Gradient Boosting	`n_estimators` = 500, `learning_rate` = 0.05, `n_jobs` = 1
Random Forest	RandomForestRegressor	`n_estimators` = 100, `n_jobs` = 4
DNN (MLP)	Multi-Layer Perceptron	`hidden_layer_sizes` = (64, 32), `max_iter` = 500, `early_stopping` = True
SVR	Support Vector Regression	`C` = 1.0, `epsilon` = 0.1, `kernel` = RBF
Kriging (GP)	GaussianProcessRegressor	`kernel` = $C(1.0) \times RBF(1.0)$, `n_restarts_optimizer` = 0

A. Global Feature Importance

The SHAP summary plot (Fig. S2) visualizes the global importance and directional impact of each design variable. The feature contribution distributions demonstrate that the intrinsic decision logic of the surrogate model strictly conforms to fundamental physical energy balance rules. For instance, the model accurately captures the monotonic decreasing effect of renewable energy penetration on operational costs. Furthermore, it explicitly reflects the penalty effect associated with diesel generators: expanding diesel generator capacity strictly drives up fixed operation and maintenance costs and fuel dependencies.

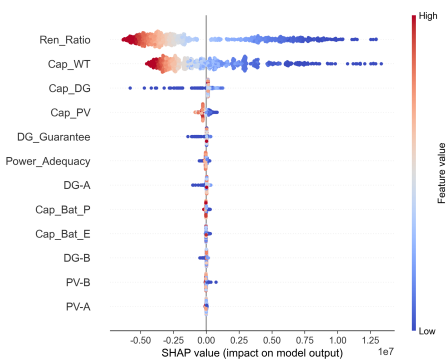

Fig. S2. SHAP summary plot detailing the directional impact of microgrid design features on operational cost.

B. Localized Interaction Effects

To further dissect the model's physical fidelity in capturing complex multi-energy synergies, the SHAP dependence plot (Fig. S3) maps localized interaction effects between specific equipment capacities. As illustrated, the surrogate successfully reconstructs the non-linear synergistic cost-reduction mechanism triggered by varying combinations of wind and solar capacity configurations, particularly under high-penetration scenarios. This interpretability analysis provides closed-loop confirmation of the surrogate model's engineering robustness and physical validity, ensuring that the surrogate-assisted optimization is guided by mathematically sound and physically realistic gradients.

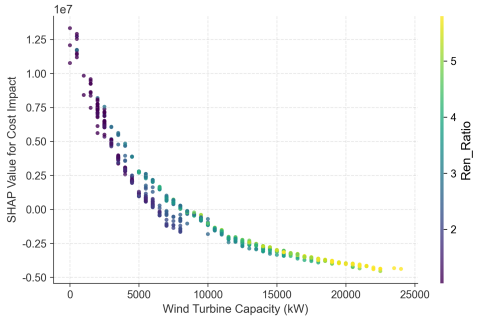

Fig. S3. SHAP dependence plot revealing the non-linear synergy between wind turbine capacity and photovoltaic penetration.